An algebraic escape from exhaustive search

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Reading conventions: [k]={1,...,k}; all logarithms are base two, except ln and exp. $\mathbb{Q}$ is the rational field. #F counts assignments on all declared variables. A polynomial with subscript parameters may have degree and constants depending on those fixed parameters. The source manuscripts report an ETH refutation; this handout explains the fixed-parameter direct SAT argument and is not an independent formal audit.

1. Target theorem and quantifier order

For every real ε > 0 there exist a deterministic algorithm Aε and positive constants Cε,cε such that, for every explicitly encoded 3-CNF F with n declared variables and m clauses, Aε decides satisfiability within the displayed number of bit operations. It suffices to construct Aε for positive rational ε: choose a smaller rational for any real target. All choices precede the input. We use the deterministic positive-rate formulation of ETH. This presentation proves that formulation false; it does not derive the optimized logarithmic runtime bound.

$$T_{A_\varepsilon}(F) \le C_\varepsilon (n+m+1)^{c_\varepsilon} 2^{\varepsilon n}$$

$$\varepsilon \;\to\; \eta = \varepsilon/2 \;\to\; C_\eta \;\to\; B_\eta = 1 + 3C_\eta \;\to\; s$$

Derived at the end from Lemmas 2–10. Runtime refers to bit operations, including reading the input.

2. Decision sparsification

For every fixed η > 0 there are a constant Cη and a deterministic algorithm which, on every n-variable 3-CNF F with m listed clauses, produces 3-CNFs Fi on the same declared variables, indexed by 1 ≤ i ≤ L, with L ≤ 2^(ηn), each containing at most Cη n clauses. Their disjunction equals F as a Boolean function. Total time is at most 2^(ηn)(n+m+1) raised to a constant depending on η. Empty clauses and empty conjunctions are handled directly. The satisfying sets need not be disjoint; the final algorithm takes an OR and does not sum branch counts.

$$F \equiv \bigvee_{i=1}^{L} F_i, \qquad L \le 2^{\eta n}, \qquad m_i \le C_\eta n$$

Stated without proof. Standard imported lemma: Impagliazzo–Paturi–Zane; Kintali Lemma 1.2.

3. Bounded occurrences and normalized factors

For every 3-CNF with n declared variables and m clauses, a polynomial-time copying reduction gives an equisatisfiable 3-CNF with N ≤ n+3m variables and at most 4m clauses. Every variable occurs in at most three clauses after removing tautologies and duplicate literals. For a variable with d ≥ 2 occurrences, use d distinct copies and add the implication cycle below. Every satisfying original assignment has a unique extension. Unused variables are retained. After removing empty-clause terminal cases, give every unused variable a dummy constant-one factor on its singleton scope. With factor set $\mathcal{F}$ and nonempty scopes Ie ⊆ [N], define M as the total incidence count. The bounds below hold. The case N=0 is decided directly.

$$\bigwedge_{j=1}^{d} (\neg x_j \vee x_{j+1}), \qquad x_{d+1} = x_1$$

$$|I_e| \le 3, \quad 1 \le |\{e : v \in I_e\}| \le 3, \quad N \le M := \sum_e |I_e| \le 3N$$

Occurrence cycle proved in the talk. Normalization and incidence bookkeeping shown. Kintali Lemma 1.3 and §2.1.

4. Deformation contract and recovery lemma

Fix integers r ≥ 2 and Q,R,D ≥ 1 and rational τ > 0, independent of N. For every normalized factor system of Lemma 3, suppose the following data are supplied exactly. Each incidence (e,v) has a homogeneous linear form Le,v(T;X) with coefficients in ℚ[T], degree in T at most R, and Le,v(0;X)=Xv identically. At every integer t ∈ [Q] there is a partition [N]=⊔g St,g with g ∈ [r] and |St,g| ≤ 2N/r; a factor assignment σt:ℱ→[r]; and integer measurements each supported in one St,g. There are at most 2τN measurements in total at each node, each with coefficient ℓ_1 norm ≤ D. Every Le,v(t;X) has a supplied rational linear representation using variables in St,σt(e) and these measurements. Then exact counting uses at most the displayed number of rational arithmetic operations. The polynomial may depend on all fixed parameters. It also covers q ≥ N via the minimum in the summation.

$$q = \lfloor 3NR/Q \rfloor, \quad J = q + 1$$

$$\mathrm{poly}(N)\left(\sum_{j=0}^{\min(q,N)} \binom{N}{j} 2^j\right)(2D+1)^{2\tau N} 2^{2N/r}$$

STATED general recovery bound; group separation and the small-support mechanism are demonstrated in the revised video. Full generic identity is given here. Source: Kintali Proposition 2.1.

5. Group conditioning identity

Fix one node t and suppress it. Write xg for the bits in Sg and Cg for the integer measurement matrix of that group, with cg rows. Thus Σg cg ≤ 2τN. Let y=(y1,...,yr), where yg ranges over the integer box [−D,D]^cg. In the node representation of each input replace Cg xg by yg; the resulting affine expression is λe,v(xσ(e);y). For every integer J ≥ 1, every family of fixed corrections be,v(z) ∈ ℚ[z]/(z^J) independent of x, and every E ⊆ [N], define Ag(xg;y,z) as the product of the factors assigned to g evaluated on λe,v+be,v. Then the sum of the full product over Boolean x with xi=1 for i∈E equals the sum over y of the product, over g, of group sums constrained by Cg xg=yg and xi=1 for i∈E∩Sg. Each assignment has one measurement tuple; infeasible tuples contribute zero. All constraints separate by group. Negative weights are allowed.

$$\sum_{x\,:\,x_i=1\ (i\in E)} \prod_{e} \Phi_e(\lambda_{e,v} + b_{e,v}) = \sum_{y} \prod_{g} \sum_{x_g\,:\,C_g x_g = y_g,\ x_i=1\ (i\in E\cap S_g)} A_g(x_g; y, z)$$

Finite distributivity over the truncated polynomial ring. Shown in the talk. The vector of group contributions, not merely their sum, is conditioned.

6. Taylor supports and Hermite uniqueness

Use the notation and assumptions of Lemma 4. Set H(T)=Σx∈{0,1}^N ∏e Φe((Le,v(T;x))v∈Ie); then H(0)=#F and deg H ≤ MR ≤ 3NR. For fixed node t and fixed Boolean base x, introduce an independent formal vector X and define Δe,v(z;X)=Le,v(t+z;X)−Le,v(t;X). Let Gx(X,z) be the factor product with inputs Le,v(t;x)+Δe,v(z;X). Since every Δ is linear in X and divisible by z, degX[z^j]Gx≤j. Boolean multilinearization modulo z^J therefore uses only supports E⊆[N] of size ≤q. For 1U the Boolean indicator vector of U, the displayed identity computes the Taylor record using fixed corrections Δ(z;1U). E fixes actual bits; U fixes correction inputs. Finally, the Q Taylor records through degree J−1 determine H: a difference is divisible by ∏t∈[Q](T−t)^J, whose degree is QJ > deg H.

$$H(t+z) \equiv \sum_{|E| \le q} \sum_{U \subseteq E} (-1)^{|E|-|U|} \sum_{x\,:\,x_i = 1\ (i \in E)} G_x(\mathbf{1}_U, z) \pmod{z^J}$$

The video proves Hermite uniqueness and illustrates support extraction at second order. This is the generic identity. Coefficients are H^(j)(t)/j!, not raw derivatives. Kintali §2.3.

7. Construction lemma

Let r,R,Q be integers with r ≥ 2, R ≥ 32, Q ≥ 6R. Let τ > 0 be rational and set D=r^(2Q). For every normalized factor system of Lemma 3 satisfying the displayed size and tail hypotheses, the complete data of Lemma 4 can be constructed deterministically in polynomial bit time in N, for these fixed parameters. The replacement degree is at most R−1. There is no claim that this polynomial-time bound is uniform in r,Q,R,τ.

$$12D^2 \le M, \quad 2Qr^2 \le N, \quad 2e^{-R^2/(8Q)} \le \tau/(3r)$$

STATED spectral and placement input. The video derives the root-filter deformation from these guarantees and explains exact measurement accounting. Source: Kintali Proposition 3.1.

8. Spectral correction and sparse disagreement

Under Lemma 7 parameters, let $\mathcal{B}=([r]\times[r])^Q$, so $|\mathcal{B}|=D$. At coordinate u write its schedules as $a_u,d_u\in[r]^Q$. For $t\in[Q]$, Π_t swaps the two symbols in pair t, acting on $\mathbb{Q}^{\mathcal{B}}$. Put $V=\prod_t\Pi_t$ and $W=(1/2)\Sigma_t(I-\Pi_t)$. Their common orthogonal tensor basis Ψ has entries in $\{-1,0,1\}$; if ψ has $j(\psi)$ antisymmetric tensor factors, $W\psi=j(\psi)\psi$ and $V\psi=(-1)^{j(\psi)}\psi$. For $p=(r-1)/(2r)$, $\mu=Qp$, $b=\lfloor\mu-(R-1)/2\rfloor$, put $I_0=\{b,\dots,b+R-1\}$. This lies in $[0,Q]$ and contains all integers within $R/4$ of μ. Choose the unique rational f of degree $\le R-1$ with $f(j)=(-1)^j$ on I_0. Let $K=Vf(W)$. It fixes all basis vectors with $j\in I_0$. The exceptional set Ψ_{bad} consists of the rest and has size $c\le 2D\exp(-R^2/(8Q))$, by the binomial tail bound. The rows ψ^{T} for $\psi\in\Psi_{bad}$ are integer measurements of norm $\le D$ and reconstruct $I-K$ by orthogonal expansion. For $E_{uw}=\{t: a_w(t)\neq d_u(t)\}$, $K[u,w]\neq 0$ implies $|E_{uw}|\le R-1$. Define $\varphi_{uw}(T)=\prod_{t\in E_{uw}}(1-T/t)$. The form P_u below equals ζ_u at zero, has degree $\le R-1$, and at node t uses only coordinates w with $a_w(t)=d_u(t)$, plus the exceptional measurements.

$$\beta_{u,\psi}=\frac{(1-(-1)^{j(\psi)}f(j(\psi)))\psi_u}{\langle\psi,\psi\rangle}$$

$$P_u(T;\zeta)=\sum_{\psi\in\Psi_{\mathrm{bad}}}\beta_{u,\psi}\langle\psi,\zeta\rangle+\sum_{w\,:\,K[u,w]\neq 0}K[u,w]\phi_{u,w}(T)\zeta_w$$

Spectral lemma STATED in the video. This companion supplies the central-level construction and coefficient formula; see the additional spectral derivation below and Kintali Lemmas 3.2–3.3.

9. Deterministic placement and measurement accounting

For every normalized factor system satisfying $12D^2 \le M$ and $2Qr^2 \le N$, schedules $a_{v,d_e} \in [r]^Q$ can be chosen deterministically in polynomial time for fixed r, Q so that every incidence label (a_{v,d_e}) has load $c_u \le 2M/D$ and every actual group $S_{t,g} = \{v : a_v(t) = g\}$ has size $\le 2N/r$. One proof uses the sum-of-squared-load potential Φ and group-size potential Ξ, with $E\Phi \le 6M$ and $E\Xi \le QN$. Conditional expectations give $\Phi \le 12M$ and $\Xi \le 2QN$. Put $h = \max_u c_u$ copies of the auxiliary array, one incidence per slot per copy; unused slots store zero. A slot containing X_v always denotes the same original bit, across all copies. For incidence (e,v) in copy j set $L_{e,v} = P_{u(e,v)}(T; \zeta^{(j)}(X))$. Assign factor e to $d_e(t)$. Restrict each exceptional measurement in each copy to each actual group. Norms remain $\le D$, including after merging repeated variables. The split measurement count obeys the displayed inequality, supplying the contract.

$$c \le D\tau/(3r), \qquad hD \le 2M \le 6N, \qquad rhc \le hD\tau/3 \le 2\tau N$$

Placement guarantee STATED in the revised video; its formal hypotheses and construction accounting are included here. Kintali Lemma 3.4 and §3.3.

10. Parameter choice, bit complexity, and conclusion

For every fixed integer s≥64 choose r=s, Q=s⁴, R=s³, τ=s⁻⁶ and D=s^(2s⁴). For N≥N0(s)=max{12D²,2Qr²}, Lemma 7 applies because N≤M and 2exp(−s²/8)≤1/(3s⁷). Below N0(s), exhaustive search is absorbed into a constant depending on s. Then q=⌊3N/s⌋. With H2 the binary entropy, support enumeration is at most (N+1)2^(N(H2(3/s)+3/s)). Adding group and measurement rates gives as below. The inequality H2(p)≤p log2(e/p) and log2(2D+1)≤2+log2 D show as=O(log s/s)→0. Exact-arithmetic lemma for this construction: grouped sums and Taylor data use O_s(N)-bit numerators and denominators; a common denominator is Bs^(3N) for a fixed integer Bs. The Hermite system has O_s(N) rows, and fraction-free elimination uses O_s(N²+N log(N+1))-bit integers, so bit costs add only a polynomial factor. Hence the bounded-occurrence counter takes poly_s(N+m+1)2^(asN) bit operations. For the original target fix η=ε/2, obtain Bη=1+3Cη, then choose s with asBη≤ε/2. Process every sparsification branch and accept if one reduced count is positive. This proves Lemma 1. We do not claim a polynomial-space implementation here.

$$a_s = H_2(3/s) + 5/s + 2s^{-6}\log_2(2s^{2s^4} + 1)$$

$$2^{\eta n}\, 2^{a_s B_\eta n} \le 2^{\varepsilon n}$$

Rates derived. Exact arithmetic lemma stated with explanation; detailed proof in Kintali §4.1 and Appendix B. Fixed-parameter conclusion only.

Spectral construction: supplementary derivation

This derivation is supplementary reading, not a compressed portion of the narrated video. It supplies the details behind the stated spectral input. For one pair of r-valued labels, use coordinate vectors eab. The swap has the orthogonal eigenbasis eaa, eab+eba, and eab−eba for a<b. There are r(r+1)/2 positive directions and r(r−1)/2 negative directions. Tensor these bases across Q positions. A uniformly counted basis direction has j negative factors distributed as Bin(Q,(r−1)/(2r)). This counts dimensions, not Boolean assignments.

On a direction with j negative factors, W has eigenvalue j and V has eigenvalue (−1)^j. Interpolate f(j)=(−1)^j on R consecutive central levels, with degree at most R−1. Thus K=Vf(W) equals identity on these central directions. The remaining dimension is at most 2D exp(−R²/(8Q)) by the binomial tail bound. Orthogonal expansion gives the coefficient formula in Lemma 8, using integer basis rows of norm at most D. Every exceptional direction is retained.

The other useful property comes from degree. W is a linear combination of identity and individual commuting swaps. A degree-d term contains at most d swaps. After multiplication by V, repeated swaps cancel, leaving at most d omitted swaps relative to the full product. An entry can therefore mismatch its assigned group at at most d nodes. Multiplying each entry by its mismatch-root polynomial yields the deformation described in the video.

For a small illustration take Q=3 and f(Y)=2Y−3, matching parity at j=1,2. Then f(W)=−Π1−Π2−Π3 and K=−Π2Π3−Π1Π3−Π1Π2. Its eigenvalues at j=0,1,2,3 are −3,1,1,−3. Each term omits one swap, hence at most one mismatch. This toy is outside the quantitative construction hypotheses and does not establish a fast algorithm or a small exceptional fraction.

Source map and scope

Kintali: Main proof route; precise fixed-parameter recovery, spectral construction, schedules, and bit arithmetic.

https://shivakintali.github.io/papers/ETH.pdf

Sellke: Visual motivation: group separation, counting deformation, swaps, and interpolation.

https://msellke.com/Files/ETH-False-Slides.pdf

O’Donnell: Cross-check of the parallel matrix/permanent route; it is not used as a black-box reduction in this talk.

https://www.cs.cmu.edu/~odonnell/some_eth_false_exposition.pdf

OpenAI manuscript: Original reported result; its sharper runtime and general counting/space claims are outside this presentation.

https://raw.githubusercontent.com/openai/math/main/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026/paper.pdf

Community examination: Context and links. Examination organizers distinguish organizing scrutiny from certifying a result.

https://ethdiscussion.blog/

The presentation proves an arbitrarily small deterministic exponential rate for decision using a bounded-occurrence exact counter. It does not prove the optimized logarithmic exponent, polynomial space, or general exact counting by ordinary decision sparsification. No screenshots, source slide artwork, or source narration are reused.